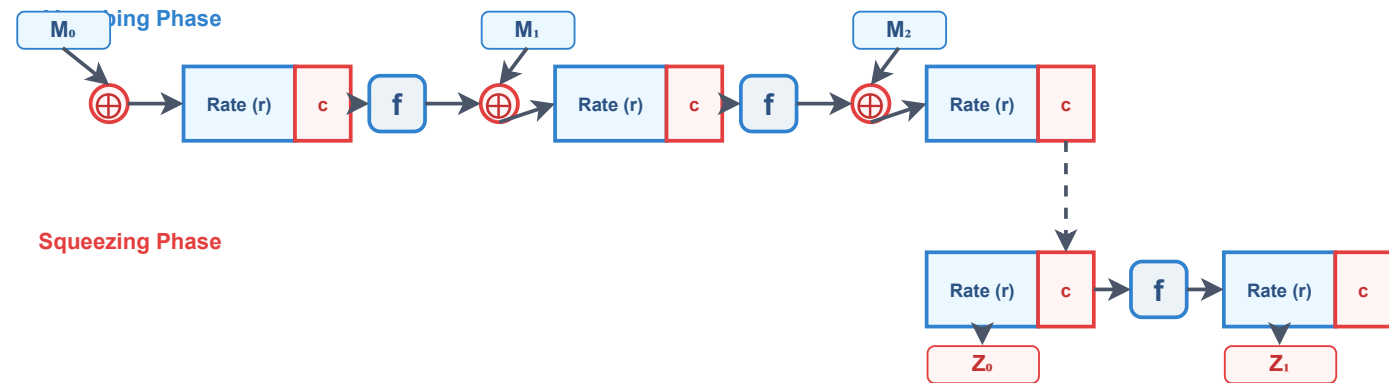


SHA-3 256 Summary Diagram

Names/ Abdulmajeed Alotaibi
Meshal Alossaimi
Tariq Alawni

1) Sponge Construction: Absorbing & Squeezing Phases



2) Message Padding (pad10*1)

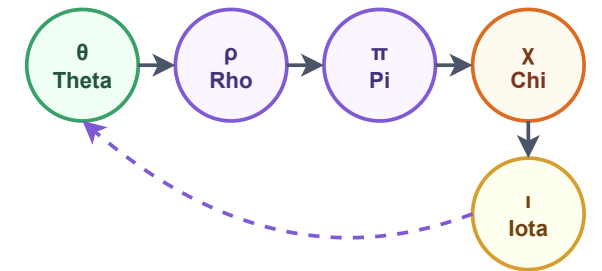
Original Message:
"AbdulmajeedMeshalTariq"
(21 bytes = 168 bits)

Padding Rule:
1. Append bit '1' (domain 01)
2. Append zeros
3. Append bit '1'

Byte View:
[Msg] 0x06 [0x00...] 0x80

Final Block: 136 bytes (1088 bits)

3) Keccak-f[1600] Permutation Cycle



Executed 24 Rounds ($i = 0$ to 23)

4) Theta (θ) — Column Parity Diffusion

Vector	Meaning	Boolean Formula
$C[x]$	Column parity	$A[x,0] \oplus A[x,1] \oplus A[x,2] \oplus A[x,3] \oplus A[x,4]$
$D[x]$	Column mask	$C[x-1] \oplus \text{ROTL64}(C[x+1], 1)$
$A'[x,y]$	Updated lane	$A[x,y] \oplus D[x]$

Same $D[x]$ is XORed into all five lanes of column x . This creates column-wide diffusion.

5) Rho (ρ) Offsets & Pi (π) Mapping

ROTL64 Offsets $r[x,y]$ in bits					
$x \backslash y$	0	1	2	3	4
0	0	36	3	41	18
1	1	44	10	45	2
2	62	6	43	15	61
3	28	55	25	21	56
4	27	20	39	8	14

Pi Mapping	
Old	New
(0,0)	(0,0)
(1,0)	(0,2)
(2,0)	(0,4)
(3,0)	(0,1)
(4,0)	(0,3)

$$X = y$$
$$Y = (2x + 3y) \bmod 5$$

6) Chi (χ) — Non-Linear Row Mixing

$$A_\chi[x,y] = B[x,y] \oplus ((\neg B[x+1,y]) \wedge B[x+2,y])$$

Current bit	Next bit	After next	Effect
bit	0	0	bit $\oplus$ 0
bit	0	1	bit $\oplus$ 1
bit	1	0	bit $\oplus$ 0
bit	1	1	bit $\oplus$ 0

Only when the next bit is 0 and the after-next bit is 1 does Chi flip the current bit.

7) Iota (ι) — Round Constant Injection

$$A'[0,0] = A_\chi[0,0] \oplus RC[i]$$

Round i	$RC[i]$ — 64-bit hexadecimal
0	0x0000000000000001
1	0x0000000000000802
2	0x800000000000080A
...	...
23	0x80000000080008008

Purpose: break symmetry so every round is not structurally identical.

8) Architectural Specifications & Hardware Implementation

Parameter	Value
Digest size	256 bits = 32 bytes
State matrix	1600 bits = 25 lanes $\times$ 64 bits
Rate / Capacity	$r = 1088$ bits $c = 512$ bits
Rounds	24 cycles (Keccak-f[1600])

ASIC / FPGA Implementation Notes

Rho / Pi: Rotations and lane movements are mainly fixed wiring/permutation. They do not require arithmetic blocks, but physical routing still has area and delay.

Throughput: Iterative cores reuse one round datapath (24 cycles per block). Unrolled or pipelined cores trade more area for higher throughput (1 block/cycle).

Area vs Speed: The key hardware design challenge is balancing the silicon footprint against the required hashing speed for network appliances.